



Report Of Machine Learning Project

**Submitted
To**

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Roll Number of Student: CS-2341589

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Section: 4CSE14

Batch: 2023-2027

CANDIDATE'S DECLARATION

I, **SHUBHAM KUMAR**, do hereby declare that the internship report titled "**Machine Learning Project**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: SHUBHAM KUMAR

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at **Pratinik Infotech**. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to my mentors at Pratinik Infotech for the opportunity and support extended to me during the internship. I would also like to thank my faculty at IILM University for their continuous guidance and encouragement throughout this internship.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: SHUBHAM KUMAR

Roll No.: CS-2341589

INTERNSHIP COMPLETION CERTIFICATE



Internship Completion Certificate issued by Pratinik Infotech

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Chapter 4: Project Description

4.1 Introduction

This report presents an account of the internship undertaken at Pratinik Infotech as a Machine Learning Intern, over a duration of two months, from 10-06-2026 to 09-08-2026. The internship provided hands-on exposure to machine learning development, working alongside experienced mentors on real project tasks combined with structured technical training. Program highlights included hands-on experience in exploratory data analysis, predictive modeling, and model deployment.

4.2 Organization Profile

Pratinik Infotech is a technology-driven organization that emphasizes hands-on, real-world work for its interns. Every internship at Pratinik Infotech is structured around practical project tasks and dedicated mentorship from experienced engineers, with the objective of building both technical competence and professional discipline in its interns.

4.3 Problem Statement

Organizations increasingly require data-driven systems that are reliable, maintainable, and capable of generating actionable insights. The project undertaken during this internship addressed the need for a predictive machine learning pipeline that could manage core data workflows, apply robust statistical algorithms, and output reliable forecasts to streamline decision-making processes.

4.4 Project Objectives

- To gain practical exposure to machine learning pipelines in a professional, project-driven environment.
- To design and develop data preprocessing, training, and evaluation scripts.
- To apply structured software development practices, including version control and documentation.
- To implement core models such as predictive classifiers and regressors.

4.5 Scope of the Project

The scope of the project covers the design and development of a machine learning workflow, including data cleaning, feature engineering, model selection, and performance evaluation within the two-month internship period.

4.6 Technologies and Tools Used

- **Programming Language:** Python 3.x
- **Data Processing:** Pandas, NumPy
- **Machine Learning:** Scikit-Learn, TensorFlow
- **Tools:** Jupyter Notebook, Git & GitHub, VS Code

4.7 System Architecture

The application follows a standard machine learning pipeline architecture consisting of a data ingestion layer, an exploratory analysis module, a model training module, and an evaluation layer. This separation of concerns makes the system easier to maintain, test, and extend.

4.8 Methodology

The internship followed a structured methodology consisting of onboarding, training modules, project tasks, and periodic evaluations. Development proceeded through requirement analysis, data gathering, incremental scripting of model components, and rigorous testing against validation datasets. Throughout the program, the intern focused on problem-solving within a standard data science workflow.

4.9 Expected Outcomes

Through this internship, the following outcomes were achieved: a working understanding of core Python development concepts and machine learning algorithms; practical experience solving data problems within a professional workflow; and professional growth through mentorship and periodic assessment.

4.10 Certificates of Completion and Other Communication Mail Proof

The Internship Completion Certificate issued by Pratinik Infotech is provided in the front matter of this report. The offer of internship confirming the role, duration, and terms of the internship is attached below as supporting communication proof.

Pratinik Infotech

Offer of Internship

Dear Shubham Kumar,

We are pleased to offer you the opportunity to join Pratinik Infotech as a Machine Learning Intern. After reviewing your qualifications, we are confident in your potential and excited about working together to support your professional growth.

Internship Details

Position: Machine Learning Intern

Duration: 2 Months (10-06-2026 to 09-08-2026).

Internship Program Overview: The Internship program is designed to provide hands-on experience in Machine Learning through real-world projects, mentorship, and assessments.

Conditions of Internship

1. **Training:** Interns must complete the 2 Month program, which includes training modules and project tasks.
2. **Performance Evaluation:** Interns will be assessed based on their achievements during the internship.
3. **Confidentiality Agreement:** Interns are required to sign a confidentiality agreement to protect the company's proprietary information.

We look forward to having you on board and helping you build your expertise during this internship.

Once again, congratulations on your selection!

Muskan Sahoo

HR Executive

Pratinik Infotech

Offer of Internship - Pratinik Infotech

Chapter 5: Bibliography/References

1. Python Software Foundation, "Python 3 Documentation," docs.python.org, 2026.
[Online].
2. Scikit-Learn Developers, "Machine Learning in Python," scikit-learn.org, 2026.
[Online].
3. Pandas Development Team, "Pandas Documentation," pandas.pydata.org, 2026.
[Online].
4. Pratinik Infotech, "Internship Onboarding Documentation," Pratinik Infotech, 2026.
5. Pratinik Infotech, "Offer of Internship - Shubham Kumar," Pratinik Infotech, 2026.